

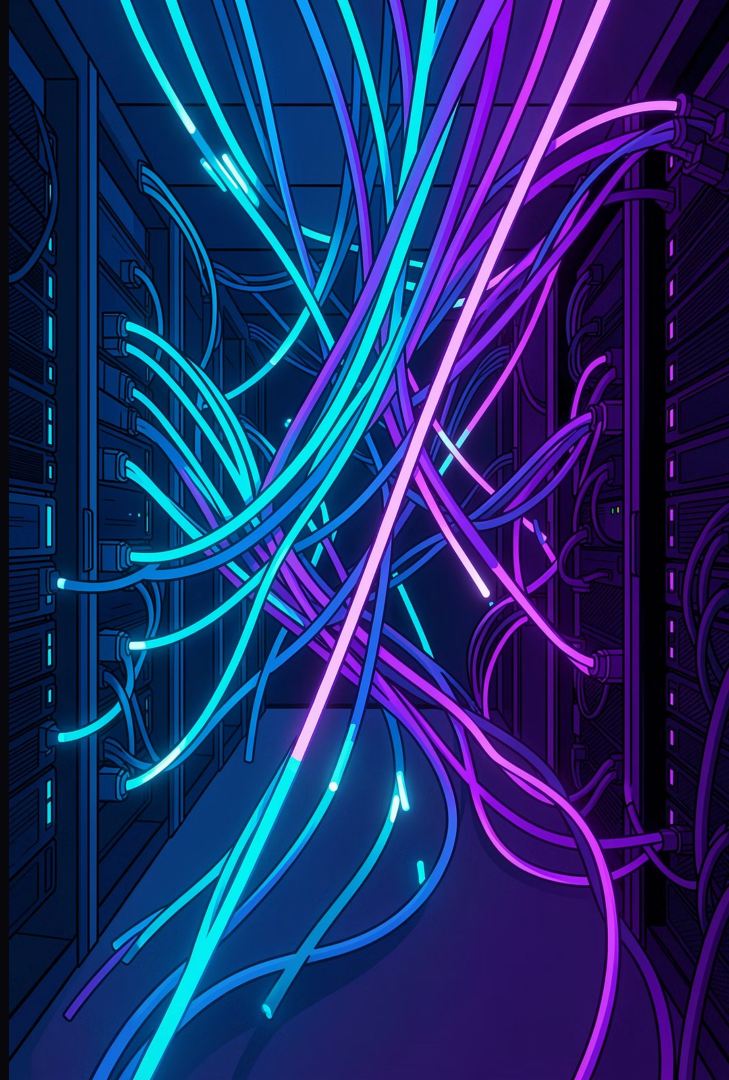
Telecom Customer Churn Prediction

A machine learning project designed to predict which telecom customers are likely to discontinue services — enabling proactive retention strategies and protecting business revenue.

MACHINE LEARNING

CLASSIFICATION

BUSINESS ANALYTICS





Project Objective

The core goal of this project is to **predict telecom customer churn** and identify customers who are likely to discontinue telecom services. By building a reliable classification model, the project helps telecom businesses take proactive action — targeting high-risk customers before they leave, improving retention rates, and protecting revenue streams.

Predict

Identify customers at risk of churn before they leave.

Retain

Enable targeted campaigns to improve customer retention.

Protect

Safeguard revenue by reducing avoidable customer loss.

Dataset & Preprocessing

A robust ML pipeline begins with clean, well-prepared data. The following preprocessing steps were applied to the telecom customer dataset to ensure high-quality inputs for modeling.

01

Load & Inspect

Loaded the telecom customer data and performed initial inspection to understand structure, data types, and potential issues.

03

Feature Engineering

Conducted feature engineering to create or transform variables, improving the model's predictive performance.

02

Clean & Handle Missing Values

Performed data cleaning and addressed missing values to ensure data completeness and reliability.

04

Encode & Standardize

Encoded categorical variables using suitable encoding techniques and standardized numerical features where required for consistent model input.



Exploratory Data Analysis (EDA)

EDA was used to explore the dataset, understand customer behavior patterns, and uncover relationships between features and churn. The analysis covered customer demographics and service usage patterns, helping to identify trends, class imbalance, and churn-related behavior that informed feature selection and model design.



Customer Demographics

Analyzed age and salary distributions to understand the demographic profile of customers and how these attributes relate to churn risk.



Service Usage Patterns

Examined call activity, SMS usage, and data consumption to identify behavioral patterns that distinguish churning customers from loyal ones.

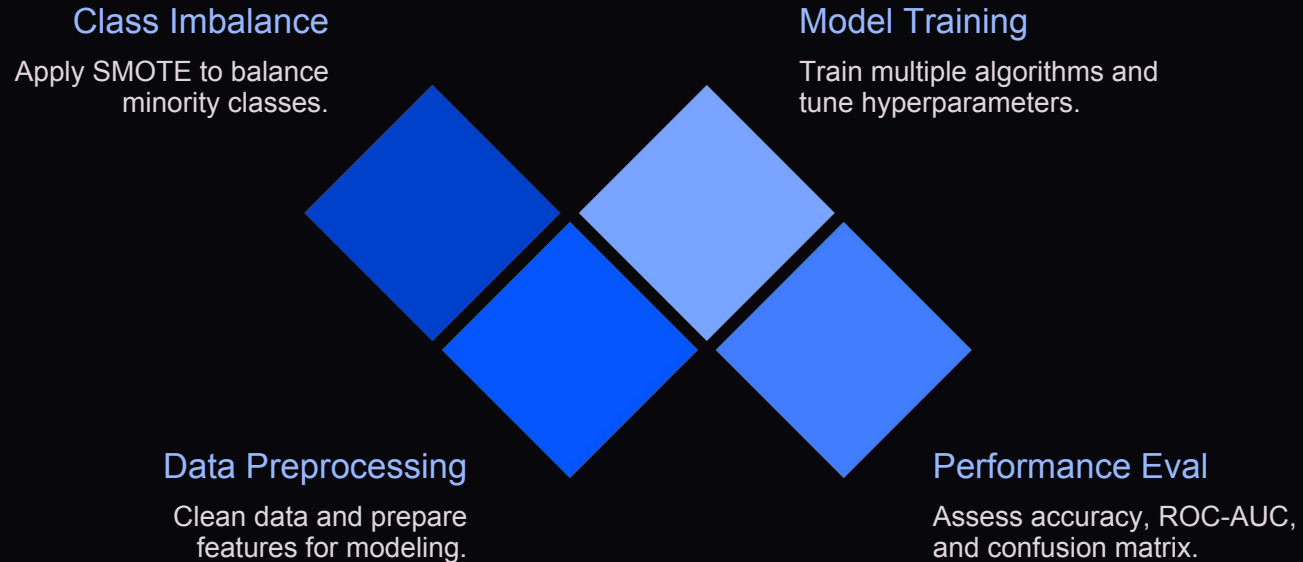


Trends & Relationships

Identified trends and feature relationships across the dataset, including the class imbalance between churned and retained customers.

Machine Learning Pipeline

The project followed a structured ML pipeline — from raw data to a trained, evaluated model — ensuring reproducibility and reliable performance measurement.



Each stage of the pipeline builds on the previous one, ensuring that data is properly prepared, class imbalance is addressed, models are trained using multiple techniques, and performance is measured with appropriate metrics to select the best solution.



Handling Class Imbalance with SMOTE

Churn datasets are often **imbalanced** — far fewer customers churn than stay. Training on imbalanced data can bias the model toward the majority class, reducing its ability to detect at-risk customers. To address this, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied during preprocessing.

What is SMOTE?

SMOTE generates synthetic examples of the minority class (churning customers) by interpolating between existing samples, balancing the training data without simple duplication.

Why It Matters

Balanced training data improves the model's ability to correctly identify churners, leading to better recall and more actionable predictions for retention teams.

Algorithms Used

Multiple machine learning techniques were trained and compared to find the best-performing model for churn prediction. The notebook implemented the following classification algorithms:

Logistic Regression

A baseline linear classifier used to establish a performance benchmark. Logistic regression is interpretable and efficient, making it a common starting point for binary classification problems like churn prediction.

Random Forest Classifier

An ensemble method that builds multiple decision trees and aggregates their predictions. Random Forest handles non-linear relationships well and is robust to overfitting, often delivering stronger predictive performance on tabular data.



Additional preprocessing and evaluation utilities were also used to support model training and performance measurement.

Model Evaluation Metrics

Three key metrics were used to measure and compare model performance, ensuring a comprehensive view beyond simple accuracy.



Accuracy

Measures the overall proportion of correct predictions. A useful baseline, but can be misleading on imbalanced datasets.



ROC-AUC Score

Measures the model's ability to distinguish between churners and non-churners across all classification thresholds. A score closer to 1.0 indicates excellent discriminative power.



Confusion Matrix

Breaks down predictions into True Positives, True Negatives, False Positives, and False Negatives — revealing exactly where the model succeeds and fails.

Business Impact

The final solution directly addresses a critical business challenge: **customer attrition**. By accurately predicting which customers are likely to churn, telecom companies can shift from reactive to proactive retention strategies.

→ Targeted Retention Campaigns

Marketing and customer success teams can focus resources on high-risk customers with personalized offers and interventions.

→ Reduced Customer Loss

Early identification of at-risk accounts enables timely action, reducing overall churn rates.

→ Improved Satisfaction & Revenue

Proactive engagement improves customer satisfaction while protecting recurring revenue streams.



Key Skills Demonstrated

This project showcases a comprehensive set of data science and machine learning competencies, spanning the full workflow from raw data to business-ready insights.

Data Engineering & Analysis

Python, Pandas, NumPy

Data Cleaning & Feature Engineering

Data Visualization & EDA

Machine Learning & Analytics

Classification Modeling

Model Evaluation (Accuracy, ROC-AUC, Confusion Matrix)

Business Analytics & Impact

- ✓ This end-to-end project demonstrates the ability to translate raw telecom data into actionable, revenue-protecting business intelligence using modern ML techniques.